

1. (Fall 2019, #4) Consider $f(x) = \frac{2}{3-x}$ with domain $[0, 2]$. Note that, the range of f is contained in $[0, 2]$. (You do not need to show this fact and may use it freely.) Let $u_0, u_1, u_2, \dots$ be the sequence defined by $u_0 = \frac{3}{2}$ and $u_{n+1} = f(u_n)$, for all $n \geq 0$.

- (a) Show that, for any $n \geq 0$, we have $0 \leq u_n \leq 2$. You **must** use a mathematical induction to prove this assertion.

$$u_0 = \frac{3}{2} \in [0, 2] \rightarrow \text{The domain of } f.$$

$$u_1 = f(u_0) \in [0, 2] \rightarrow \text{The range of } f.$$

$$\text{Suppose } u_{n-1} \in [0, 2] \rightarrow \text{The domain of } f.$$

$$\text{check } u_n = f(u_{n-1}) \in [0, 2] \rightarrow \text{The range of } f.$$

$$\Rightarrow \forall n \quad 0 \leq u_n \leq 2.$$

- (b) Show that the sequence $u_0, u_1, u_2, \dots$ is decreasing. You **must** use mathematical induction to prove this assertion. **Hint:** Use the fact that f is increasing.

Note that f is increase.

$$u_1 = f(u_0) = \frac{2}{3-\frac{3}{2}} = \frac{4}{3} < u_0 = \frac{3}{2}.$$

$$\text{Suppose } u_{n-1} < u_{n-2}.$$

$$\text{check } u_n = f(u_{n-1}) < f(u_{n-2}) = u_{n-1} \Rightarrow u_0, u_1, u_2, \dots \text{ is decreasing,}$$

$\downarrow$
 $f \text{ is increasing.}$

- (c) Show that the sequence $u_0, u_1, u_2, \dots$ converges to a limit l , then determine the value of l . You must carefully justify your answer.

$$u_{n+1} = f(u_n) \quad (*)$$

Since u_n is decreasing, and $u_n \in [0, 2] \quad \forall n$.
 By the convergent property of bounded monotone sequence.
 $\Rightarrow, \exists l \quad \lim_{n \rightarrow \infty} u_n = l.$

Since f is continuous, send $n \rightarrow \infty$ in $(*)$.

$$\Rightarrow l = f(l) \Rightarrow l = \frac{2}{3-l} \Rightarrow l(3-l) = 2$$

$$\Rightarrow l = 1.$$

$$\Rightarrow l^2 - 3l + 2 = 0 \Rightarrow (l-2)(l-1) = 0 \Rightarrow l = 1 \text{ or } l = 2, \text{ since } u_n \text{ is decreasing and } u_0 = \frac{3}{2} < 2.$$

2. (Fall 2019, #5c) Let $0 < r < s$ be constants. Does the sequence $\{(r^n + s^n)^{1/n}\}$ converge as $n \rightarrow \infty$? If it does converge, what is the limit?

Yes.

Consider $f(x) = (r^x + s^x)^{\frac{1}{x}} \quad x > 0.$

$$f(n) = (r^n + s^n)^{\frac{1}{n}}.$$

$\exists f \quad \lim_{x \rightarrow \infty} f(x) = L. \quad \text{Then } \lim_{n \rightarrow \infty} f(n) = L.$

check $\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} (r^x + s^x)^{\frac{1}{x}}$

$$\begin{aligned} 0 < r < s \\ &= \lim_{x \rightarrow \infty} \left[\left(\frac{r^x}{s^x} + 1 \right) \cdot s^x \right]^{\frac{1}{x}} \\ &= \lim_{x \rightarrow \infty} \left[\left(\frac{r}{s} \right)^x + 1 \right]^{\frac{1}{x}} \cdot s. \end{aligned}$$

Note that $0 < \frac{r}{s} < 1$

$$1 \leq \left[\left(\frac{r}{s} \right)^x + 1 \right] \leq 2.$$

$$\Rightarrow \left[\left(\frac{r}{s} \right)^x + 1 \right]^{\frac{1}{x}} \leq 2^{\frac{1}{x}}.$$

$\downarrow$
 $1^0 = 1$

$\downarrow$
 $2^0 = 1$

send $x \rightarrow \infty$

By squeeze principle, $\lim_{x \rightarrow \infty} \left[\left(\frac{r}{s} \right)^x + 1 \right]^{\frac{1}{x}} = 1 \Rightarrow \lim_{x \rightarrow \infty} f(x) = s. \Rightarrow \lim_{n \rightarrow \infty} f(n) = s.$

3. (Spring 2013, #5)

(a) If $a_n = \frac{\sqrt{n} \sin(n!)}{n+1}$, carefully determine if $\lim_{n \rightarrow \infty} a_n$ exists, and if so, find it.

$$\text{Note that } -\frac{1}{n} \leq \frac{-\ln}{n} \leq a_n \leq \frac{\ln}{n} \leq \frac{1}{n}.$$

$$\text{And } \lim_{n \rightarrow \infty} \frac{1}{n} = \lim_{n \rightarrow \infty} \frac{-1}{n} = 0. \text{ Then, by}$$

$$\text{squeeze principle, } \lim_{n \rightarrow \infty} a_n = 0.$$

(b) Let $c > 0$, set $b_1 = c$, $b_2 = \sqrt{c} + 1$ and in general let $b_{n+1} = \sqrt{b_n} + 1$.

i. Show that the sequence (b_n) is increasing or decreasing, depending on c .

(i)

$$b_1 = c, \quad b_2 = \sqrt{c} + 1$$

Case 1:

$$\text{If } c > \sqrt{c} + 1 \text{ i.e. } (\sqrt{c})^2 - \sqrt{c} - 1 > 0 \Rightarrow \sqrt{c} > \frac{1+\sqrt{5}}{2}.$$

We have $b_2 < b_1$. Then we claim $\{b_n\}$ is decreasing.
Show by induction. Suppose $b_n < b_{n-1}$. Then.

$$b_{n+1} = \sqrt{b_n} + 1 < \sqrt{b_{n-1}} + 1 = b_n. \Rightarrow \{b_n\} \text{ decreasing.}$$

ii. Show that (b_n) is bounded.

$$(ii) \quad \textcircled{1} \text{ If } c > \sqrt{c} + 1 \Rightarrow b_1 = c, \quad b_2 = \sqrt{c} + 1 < c$$

$$\Rightarrow b_3 = \sqrt{b_2} + 1 < \sqrt{c} + 1 < c.$$

$$\Rightarrow \forall n, \quad b_n < c. \Rightarrow 0 < b_n < c.$$

$$\textcircled{2} \text{ If } c = \sqrt{c} + 1 \Rightarrow \forall n, \quad b_n = c.$$

$$\textcircled{3} \text{ If } c < \sqrt{c} + 1 \Rightarrow \sqrt{c} < \frac{1+\sqrt{5}}{2} := c^*.$$

iii. Justify whether (b_n) converges or diverges.

$$\Rightarrow c < \frac{(1+\sqrt{5})^2}{4} = \frac{3+\sqrt{5}}{2} < 3.$$

$$\text{Then, } b_1 = c < 3$$

$$b_2 = 1 + \sqrt{c} < 1 + \sqrt{3} < 3.$$

$$b_3 = 1 + \sqrt{b_2} < 1 + \sqrt{3} < 3.$$

$$\text{We claim that } b_n < 3, \forall n.$$

iv. If (b_n) converges find its limit.

In fact, suppose $b_n < 3$. Then

$$b_{n+1} = 1 + \sqrt{b_n} < 1 + \sqrt{3} < 3.$$

$\Rightarrow$ By induction,

$$0 < b_n < 3.$$

$\Rightarrow \{b_n\}$ is always

bounded.

(iii) Since $\{b_n\}$ is monotone and bounded. we have $\{b_n\}$ converges.

$$\text{Case 2: If } c = \sqrt{c} + 1$$

$$\Rightarrow b_1 = b_2 = \dots = b_n \Rightarrow \text{Constant.}$$

$$\text{Case 3: If } c < \sqrt{c} + 1.$$

$$\Rightarrow 0 < \sqrt{c} < \frac{1+\sqrt{5}}{2}. \text{ Then } b_1 < b_2.$$

We claim in this case $\{b_n\}$ is increasing. Suppose that $b_{n-1} < b_n$.

$$\text{Then } b_n = \sqrt{b_{n-1}} + 1 < \sqrt{b_n} + 1 = b_{n+1}.$$

$$\Rightarrow \{b_n\} \text{ is increasing.}$$

$$(iv) \text{ Since } b_{n+1} = \sqrt{b_n} + 1 \text{ converges.}$$

$$\text{Let } \lim_{n \rightarrow \infty} b_n = L. \text{ Then, send } n \rightarrow \infty \text{ on both sides,}$$

$$L = \sqrt{L} + 1. \text{ since } L > 0.$$

$$\Rightarrow (\sqrt{L})^2 - \sqrt{L} - 1 = 0$$

$$\Rightarrow \sqrt{L} = \frac{1+\sqrt{5}}{2} \Rightarrow L = \frac{(1+\sqrt{5})^2}{4} = \frac{6+2\sqrt{5}}{4} = \frac{3+\sqrt{5}}{2}.$$

4. (Spring 2011, #6) In each case below, the n -th term of a sequence is given. Give reasons why the sequence diverges, or why it converges and find its limit.

(a) $a_n = \left(\frac{4n-3}{n} - \frac{3n}{n+1} \right)^{n^2+n}$

Converge.

Note that $\frac{4n-3}{n} - \frac{3n}{n+1} = 4 - \frac{3}{n} - \frac{3n}{n+1}$.

$\Rightarrow a_n = \left(1 - \frac{3}{n(n+1)} \right)^{n(n+1)}$.

as $n \uparrow$, a_n is increasing.

and for $n > 2$, $0 < a_n < 1 \Rightarrow a_n$ bounded.

$\Rightarrow \{a_n\}$ converge!

$\lim_{n \rightarrow \infty} a_n = \lim_{x \rightarrow \infty} \left(1 - \frac{3}{x} \right)^x = e^{-3}$.

In fact, $\lim_{x \rightarrow \infty} \left(1 - \frac{3}{x} \right)^x = \lim_{x \rightarrow \infty} e^{x \ln \left(1 - \frac{3}{x} \right)}$

$= \lim_{x \rightarrow \infty} e^{x \cdot \left(-\frac{3}{x} \right)} = e^{-3}$.

$= 4 - \frac{3}{n} - \frac{3(n+1)-3}{n+1}$

$= 4 - \frac{3}{n} - 3 + \frac{3}{n+1}$

$= 1 - \frac{3}{n} + \frac{3}{n+1} = 1 - \frac{3}{n(n+1)}$.

(b) $b_n = (3 + \sin(n))^{1/n}$

Converge

$2^{\frac{1}{n}} = (3-1)^{\frac{1}{n}} \leq b_n \leq (3+1)^{\frac{1}{n}} = 4^{\frac{1}{n}}$

$\downarrow n \rightarrow \infty$
1

$\downarrow n \rightarrow \infty$
1

Then by squeeze principle, $\lim_{n \rightarrow \infty} b_n = 1$.

(c) $c_n = (-1)^n (1 - (1/n))$

Diverge.

$c_{2k} = (-1)^{2k} \left(1 - \frac{1}{2k} \right) = 1 - \frac{1}{2k} \rightarrow 1$ as $k \rightarrow \infty$

$c_{2k+1} = (-1)^{2k+1} \left(1 - \frac{1}{2k+1} \right) = - \left(1 - \frac{1}{2k+1} \right) \rightarrow -1$, as $k \rightarrow \infty$.

Two subsequences have two different limits.

$\Rightarrow \{c_n\}$ diverges.

5. (Spring 2010, #6) Determine whether each of the following sequences is convergent or divergent. Find the limit if the sequence is convergent. (Remember: **You must show all your work and justify your answers to get full credit**)

✓ (a) $a_n = \frac{(3 \cos(n))^n}{5^n + 2^n}$

$$\frac{-3^n}{5^n} \leq \frac{-3^n}{5^n + 2^n} \leq a_n \leq \frac{3^n}{5^n + 2^n} \leq \frac{3^n}{5^n}.$$

Note $\lim_{n \rightarrow \infty} -\left(\frac{3}{5}\right)^n = \lim_{n \rightarrow \infty} \left(\frac{3}{5}\right)^n = 0.$

Then by squeeze principle. $\lim_{n \rightarrow \infty} a_n = 0.$

$\{a_n\}$ converges.

✓ (b) $a_n = (-1)^{n+1} \frac{4n^2 + 1}{5n - n^2}$

Diverges. Denote

$$b_n = \frac{4n^2 + 1}{5n - n^2}.$$

Note that

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{4n^2 + 1}{5n - n^2} &= \lim_{x \rightarrow \infty} \frac{4x^2 + 1}{5x - x^2} \\ &= \lim_{x \rightarrow \infty} \frac{4 + \frac{1}{x^2}}{\frac{5}{x} - 1} \\ &= -4. \end{aligned}$$

Then.

$$\begin{aligned} \lim_{k \rightarrow \infty} a_{2k} &= \lim_{k \rightarrow \infty} -b_{2k} \\ &= 4. \end{aligned}$$

$$\lim_{k \rightarrow \infty} a_{2k+1} = \lim_{k \rightarrow \infty} b_{2k+1} = -4.$$

There are two subsequences having two different limits.

$\Rightarrow \{a_n\}$ diverges.